Subsea Integrity Copilot - Customer Brief

The decision we want to improve. Which pressure alerts need a leak investigation, which indicate unreliable instruments, and where should an engineer inspect first? The proposal is evidence-backed investigation alongside existing detectors, not another alarm dashboard or autonomous safety system.

Customer and problem. NorthSea Integrity Operations is a fictional offshore gas operator. Its sensors stream pressure readings along a long pipeline network; existing anomaly detectors flag unusual values. Engineers must then compare nearby and paired instruments, source times, pipeline topology and operating records. A leak can produce related pressure changes around a section, while faulty instruments can produce scattered spikes, gaps or drift. But operating changes and shared equipment failures can also affect several sensors, and leaks and faults can coexist. An anomaly is therefore a starting point, not a diagnosis. This manual investigation concentrates work in a few experienced engineers and makes knowledge difficult to transfer. Missed leaks risk environmental releases, disruption and regulatory exposure; false diagnoses can mobilize costly vessel or ROV inspections.

Why I chose this problem. My PhD studied marine data quality, anomaly detection and translating expert requirements into reviewable software. This applies that experience to a high-consequence decision where trusting the measurements matters as much as detecting change.

What the solution does. Numerical code normalizes measurements and raises warnings without waiting for AI. Three scoped agent roles investigate sensor quality, assess event evidence and coordinate an engineer-facing response. They compare pressure histories and read incident-time records, then explain competing causes, a supported inspection interval or why one cannot be justified. A grounding reviewer checks source support; rejected reports are withheld. The engineer retains the decision.

What we will demonstrate. "Leak or faulty sensors?" Compare a leak footprint with scattered instrument faults and a mixed case. "Where should we inspect, and why?" Inspect the reported interval and measured comparisons, not an invented exact coordinate. "What supports this, and what remains unknown?" Open frozen readings, cited records, limitations and the executed investigation. The Evaluation workspace compares findings with synthetic labels authored before inference, retaining failures, misses and false alarms. Answer-quality ratings are shown separately as advisory model opinions. Saved Detection playback requires no new inference; fresh investigations expose actual model latency.

The two customer roles and objections. The **Head of Subsea Operations** owns the budget and asks whether lower investigation effort justifies integration, inference and review costs without increasing operational risk. The **Lead Subsea Integrity & Instrumentation Engineer** asks whether conclusions survive missing data, inconsistent units, conflicting documents, shared failures and reproduction from source evidence. They have not signed because a cheaper fixed workflow may suffice, AI can be wrong, and data movement and accountability need approval.

The offer and buying gate. Agree a proposed four-week, read-only shadow pilot on one pipeline, subject to data/security access. Establish a manual/fixed-workflow baseline, an inference spend cap and acceptance thresholds before starting. Measure engineer review time, missed/false findings, useful inspection intervals, unsupported claims, abstention and end-to-end availability/latency, including failed attempts. Expand only if evidence quality, effort and total cost justify adoption; no savings or response-time guarantee is assumed.

Assumptions and limits. Data and records are synthetic; pressure behavior is not a validated hydraulic model. Inspection intervals are priorities, not confirmed leak coordinates or dispatch authorization. Hosted inference needs approval; production connectors and expert-approved procedures require customer work. There is no independent engineering validation or field-reliability claim. Humans retain safety authority.

Demo and run instructions: github.com/ntnguyen-so/subsea-integrity-copilot.